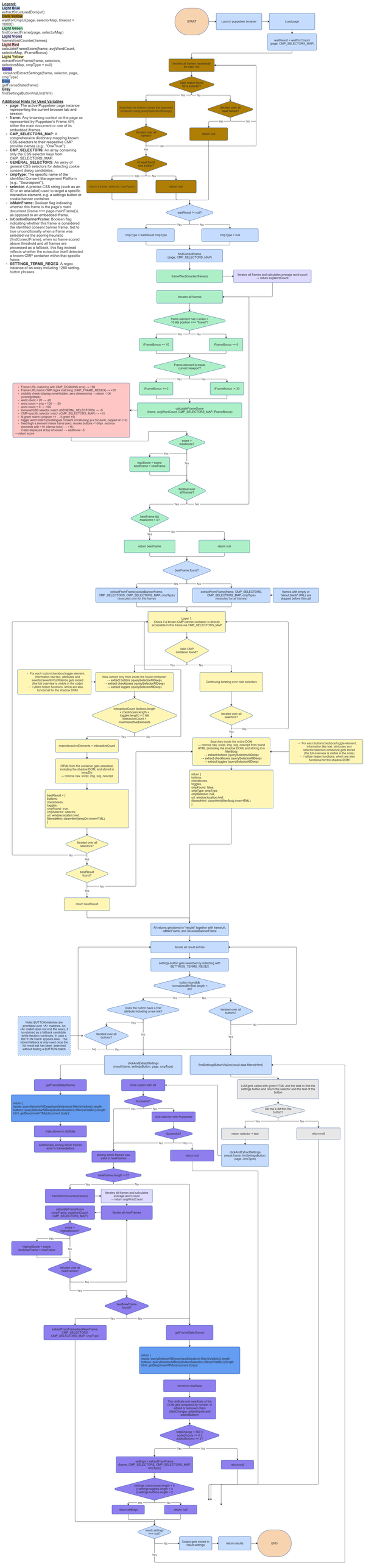

Legend:

Light Blue
extractStructuredDom(url)
Dark Yellow
waitForCmpUi(page, selectorMap, timeout = 10000)
Light Green
findCorrectFrame(page, selectorMap)
Light Violet
frameWordCounter(frames)
Light Red
calculateFrameScore(frame, avgWordCount, selectorMap, iFrameBonus)
Light Yellow
extractFromFrame(frame, selectors, selectorsMap, cmpType = null)
Violet
clickAndExtractSettings(frame, selector, page, cmpType)
Blue
getFrameState(frame)
Gray
findSettingsButtonViaLlm(html)

Additional Hints for Used Variables

- **page**: The active Puppeteer page instance representing the current browser tab and session.
- **frame:** Any browsing context on the page as represented by Puppeteer's Frame API, either the main document or one of its embedded iframes.
- **CMP_SELECTORS_MAP**: A comprehensive dictionary mapping known CSS selectors to their respective CMP provider names (e.g., "OneTrust").
- **CMP_SELECTORS**: An array containing only the CSS selector keys from CMP_SELECTORS_MAP.
- **GENERAL_SELECTORS**: An array of general CSS selectors for detecting cookie consent dialog candidates.
- **cmpType**: The specific name of the identified Consent Management Platform (e.g., "Sourcepoint").
- **selector**: A precise CSS string (such as an ID or an aria-label) used to target a specific interactive element, e.g. a settings button or cookie banner container.
- **isMainFrame:** Boolean flag indicating whether this frame is the page's main document (frame === page.mainFrame()), as opposed to an embedded iframe.
- **isCookieBannerFrame**: Boolean flag indicating whether this frame is considered the identified consent banner frame. Set to true unconditionally when a frame was selected via the scoring heuristic (findCorrectFrame); when no frame scored above threshold and all frames are processed as a fallback, this flag instead reflects whether the extraction itself detected a known CMP container within that specific frame.
- **SETTINGS_TERMS_REGEX**: A regex instance of an array including 1280 setting-button phrases.

START
Launch puppeteer browser
Load page
waitResult = waitForCmpUi (page, CMP_SELECTORS_MAP)
Iterates all frames repeatedly for max 10s
Valid match found for a selector?
Yes
No
Searches for buttons inside the assumed container using querySelectorAllDeep()
Iterated over all newFrames?
Yes
No
return null
Iterated over all frames?
No
Yes
At least found one button?
Yes
No
return { frame, selector, cmpType }
return null
waitResult != null?
Yes
No
cmpType = waitResult.cmpType
cmpType = null
findCorrectFrame (page, CMP_SELECTORS_MAP)
frameWordCounter(frames)
Iterates all frames and calculates average word count → return avgWordCount
Iterates all frames
frame element has z-index > 10 && position === "fixxed"?
Yes
No
iFrameBonus += 10
iFrameBonus += 0
Frame element is inside current viewport?
Yes
No
iFrameBonus += 0
iFrameBonus -= 30
calculateFrameScore (frame, avgWordCount, CMP_SELECTORS_MAP, iFrameBonus)
Frame URL matching with CMP_DOMAINS array → +50
Frame URL/name CMP regex matching (CMP_FRAME_REGEX) → +20
visibility check (display:none/hidden, zero dimensions) → return -100 (scoring stops)
word count < 20 → -20
word count > avg + 100 → -20
word count = 0 → -100
General CSS selector match (GENERAL_SELECTORS) → +5
CMP-specific selector match (CMP_SELECTORS_MAP) → +10
N-gram match (unigram +1 ... 5-gram +5)
trigger word match (multilingual consent vocabulary) (+2 for each, capped at +10)
fixed/high-z element inside frame (excl. revoke buttons <100px and nav elements with >10 internal links) → +15; if also displayed at top of screen → additional +5
→ return score
score > maxScore?
Yes
No
maxScore = score; bestFrame = newFrame
Iterated over all frames?
Yes
No
bestFrame && maxScore > 0?
Yes
No
return bestFrame
return null
bestFrame found?
Yes
No
extractFromFrame(cookieBannerFrame, CMP_SELECTORS, CMP_SELECTORS_MAP, cmpType) (executed only for this frame)
extractFromFrame(frame, CMP_SELECTORS, CMP_SELECTORS_MAP, cmpType) (executed for all frames)
frames with empty or "about:blank" URLs are skipped before this call
Layer 1: Check if a known CMP banner container is directly accessible in this frame via CMP_SELECTORS_MAP
Valid CMP container found?
Yes
No
For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
I utilize helper functions, which are also functional for the shadow DOM
Now extract only from inside the found container! → extract buttons (querySelectorAllDeep) → extract checkboxes (querySelectorAllDeep) → extract toggles (querySelectorAllDeep)
Continuing iterating over next selectors
interactiveCount (buttons.length + checkboxes.length + toggles.length) > 0 && interactiveCount > maxInteractiveElements
Yes
No
Iterated over all selectors?
Yes
No
maxInteractiveElements = interactiveCount
Searches inside the entire DOM → remove nav, script, img, svg, noscript from found HTML (including the shadow DOM) and storing it in filterBody → extract buttons (querySelectorAllDeep) → extract checkboxes (querySelectorAllDeep) → extract toggles (querySelectorAllDeep)
For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
I utilize helper functions, which are also functional for the shadow DOM
HTML from the container gets extracted, including the shadow DOM, and stored in tempDiv → remove nav, script, img, svg, noscript
return { buttons, checkboxes, toggles, cmpFound: false, cmpType: cmpType, cmpSelector: null, url: window.location.href, filteredHtml: cleanHtml(filterBody.innerHTML) }
bestResult = { buttons, checkboxes, toggles, cmpFound: true, cmpSelector: selector, url: window.location.href, filteredHtml: cleanHtml(tempDiv.innerHTML) }
Iterated over all selectors?
No
Yes
bestResult found?
No
Yes
return bestResult
All returns get stored in "results" together with frameUrl, isMainFrame, and isCookieBannerFrame
Iterate all result entries
settings-button gets searched by matching with SETTINGS_TERMS_REGEX
button found&& normalizedBtnText.length < 30?
Yes
No
Does the button have a href attribute including a real link?
Yes
No
Iterated over all buttons?
No
Yes
Note: BUTTON matches are prioritized over <A> matches. An <A> match does not end the searc, it is retained as a fallback candidate while iteration continues, in case a BUTTON match appears later. The stored fallback is only used once the full result set has been searched without finding a BUTTON match
Iterated over all buttons?
No
Yes
clickAndExtractSettings (result.frame, settingsButton, page, cmpType)
findSettingsButtonViaLlm(result.data.filteredHtml)
LLM gets called with given HTML and the task to find the settings button and return the selector and the text of the button
Did the LLM find the button?
Yes
No
return selector + text
return null
clickAndExtractSettings (result.frame, llmSettingsButton, page, cmpType)
getFrameState(frame)
return { inputs: querySelectorAllDeep(inputSelectors).filter(isVisible()).length buttons: querySelectorAllDeep(buttonSelectors).filter(isVisible()).length html: getDeepInnerHTML(document.body)}
Gets stored in oldState
Additionally storing which frames exist in framesBefore
Click button with JS
Sucessfull?
Yes
No
click selector with Puppeteer
Sucessfull?
Yes
No
return null
storing which frames now exist in newFrames
newFrames.length > 0?
Yes
No
frameWordCounter(frames)
Iterates all frames and calculates average word count → return avgWordCount
calculateFrameScore (newFrame, avgWordCount, CMP_SELECTORS_MAP)
Iterate all newFrames
score > highestScore?
Yes
No
highestScore = score; bestnewFrame = newFrame
Iterated over all newFrames?
No
Yes
bestNewFrame found?
Yes
No
extractFromFrame(bestNewFrame, CMP_SELECTORS, CMP_SELECTORS_MAP, cmpType)
getFrameState(frame)
return { inputs: querySelectorAllDeep(inputSelectors).filter(isVisible()).length buttons: querySelectorAllDeep(buttonSelectors).filter(isVisible()).length html: getDeepInnerHTML(document.body)}
stored in newState
The oldState and newState of the DOM ges compared by number of added or removed chars (totalChange), addedInputs and addedButtons
totalChange > 500 || addedInputs >= 0 || addedButtons >= 2?
Yes
No
settings = extractFromFrame (frame, CMP_SELECTORS, CMP_SELECTORS_MAP, cmpType)
return null
settings.checkboxes.length > 0 || settings.toggles.length > 0 || settings.buttons.length > 2
Yes
No
return settings
return null
result.settings === null?
No
Yes
Output gets stored in result.settings
return results
END